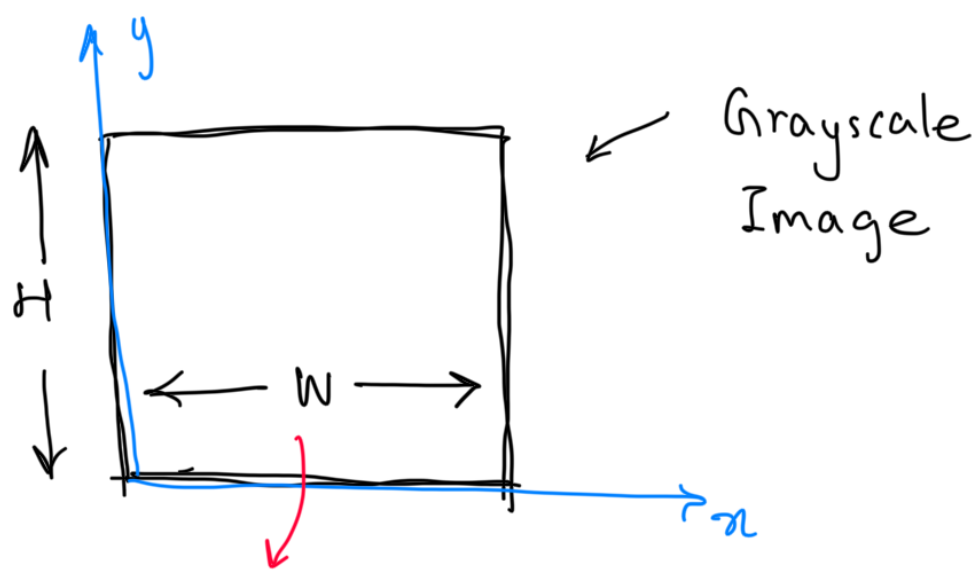




I is the black intensity $\rightarrow I(x,y)$ - the image

$\therefore x$ is from 0 to $W-1$

y is from 0 to $H-1$

also, $I(x,y)$ is in $[0, 255]$

$\uparrow$
8-bit grayscale intensity

⊛ Taking the Fourier Transform

for a continuous 2D signal,

$$F(u,v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} I(x,y) e^{-j2\pi(ux+vy)} dx dy$$

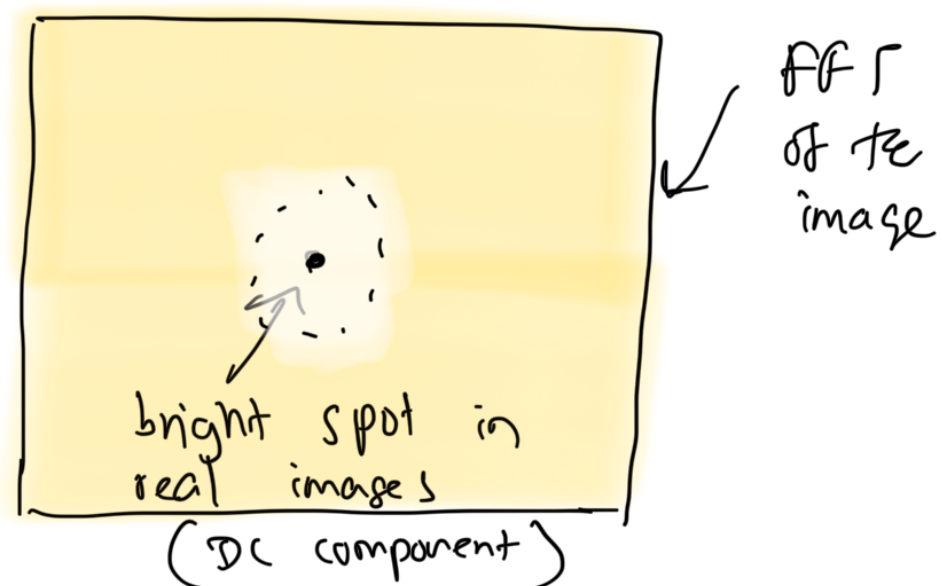
But the digital image is a discrete signal.

therefore the "discrete fourier transform of the signal (image) should be taken.

$$f[u, v] = \sum_{m=0}^{H-1} \sum_{n=0}^{W-1} I[m, n] e^{-j2\pi \left(\frac{m}{H}u + \frac{n}{W}v \right)}$$

u from 0 to $H-1$
 v from 0 to $W-1$

(*) For visualization, FFT should be shifted so that the DC component is in the middle of the plot



$$F_{\text{shift}}(u, v) = F\left(\left(u + \frac{W}{2}\right) \bmod W, \left(v + \frac{H}{2}\right) \bmod H\right)$$

in python code

$$f = \text{np.fft.fft2}(\text{img_gray.astype(np.float32)})$$

take
 numpy
 fft

convert cv2 read
 image into a float function

